Feature Engineering

P vs N

11/02/2026



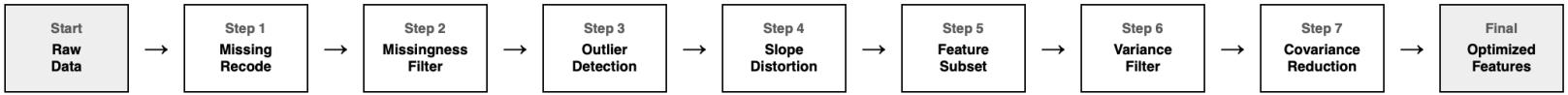
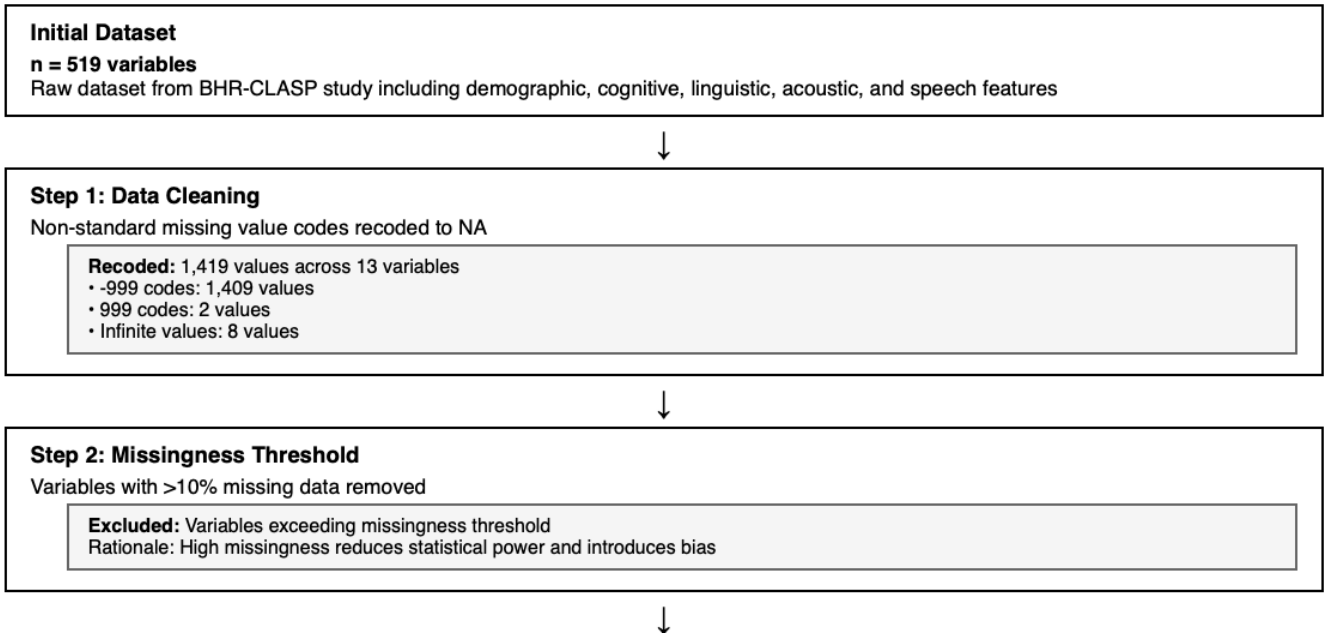

Start
Raw Data
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Step 2
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Step 3
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Step 4
Slope Distortion
Step 5
Feature Subset
Step 6
Variance Filter
Step 7
Covariance Reduction
Final
Optimized Features

Figure 1 Feature Selection Pipeline



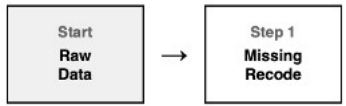


Table S1 Non-Standard Missing Value Codes in BHR Data

Variable	Data Type	Code	Count	Percentage	NaN Count
NUM_SINGLE_REPS	float64	-999.0	700	83.93	1
NUM_MULTI_REPS	float64	-999.0	700	83.93	1
THIRD_FIRST_PERSON_RATIO	float64	inf	4	0.48	13
OTAN_NOUNS	float64	-999.0	3	0.36	1
OTAN_NVAA	float64	-999.0	2	0.24	1
NOUN_VERB_RATIO	float64	inf	2	0.24	1
NOUN_PRON_RATIO	float64	inf	2	0.24	1
PTAN_NOUNS	float64	-999.0	1	0.12	1
VALENCE_NVAA	float64	-999.0	1	0.12	1
AROUSAL_NVAA	float64	-999.0	1	0.12	1
DOMINCE_NVAA	float64	-999.0	1	0.12	1
median_rxntime.flanker	float64	999.0	1	0.12	25
sarx_median_rxntime.flanker	float64	999.0	1	0.12	25
qid7_1_text	object	Unknown	1	0.12	286
qid7_1_text	object	unknown	1	0.12	286

Non-standard missing value codes identified in BHR cognition dataset (N=834 observations, 2,540 variables). The most common issue is -999.0 used as a missing indicator in NUM_SINGLE_REPS and NUM_MULTI_REPS (700 occurrences each, 83.93%). Infinity (inf) values appear in ratio variables when denominators are zero. These values should be recoded as proper NA/NaN before missingness analysis. Percentage column shows proportion of total observations. NaN Count shows number of observations already coded as proper missing values.

Table 1 Participant Demographics

Characteristic	N	%	Summary Statistics
Total Participants	**798**	**100.0**	—
Age (years)			62.4 ± 12.6 (19.0–92.0)
<50	153	18.4	—
50–64	258	31.0	—
65–79	374	44.9	—
80+	48	5.8	—
Gender			
Female	584	73.2	—
Male	197	24.7	—
Not answered	14	1.8	—
Non-binary	3	0.4	—
Education (years)			16.0 ± 2.4 (6.0–20.0)
≤12 (High school or less)	40	4.8	—
13–16 (Some college/Bachelor's)	479	57.5	—
17–20 (Graduate degree)	314	37.7	—
Race/Ethnicity			
White	678	85.0	—
Multiracial	39	4.9	—
Black/African American	34	4.3	—
Asian	26	3.3	—
Other	13	1.6	—
Native American	7	0.9	—
Pacific Islander	1	0.1	—
Visit Pattern			
Single visit	780	97.7	2022-06-27 to 2024-05-02
Multiple visits	18	2.3	Multiple dates

Demographic characteristics of participants in the BHR-CLASP study (N=798). Continuous variables presented as mean ± standard deviation (range). Categorical variables presented as frequency and percentage. Data collected between June 2022 and May 2024.

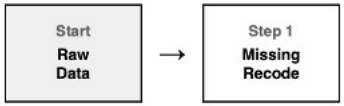


Table 2 Feature Categories in Dataset

Category	N	%	>10% Missing	Description
Total Features	**2,540**	**100.0**	**2,057 (81.0%)**	**Complete feature set in raw dataset**
Other	965	38.0	764 (79.2%)	Uncategorized variables
Medical History	1,200	47.2	1,186 (98.8%)	Clinical history variables; not used in current analysis
Acoustic/Waveform	130	5.1	0 (0.0%)	Prosodic and spectral features tested for device effects
Metadata/IDs	123	4.8	79 (64.2%)	Participant identification and device tracking
Linguistic/Speech	63	2.5	7 (11.1%)	Lexical, syntactic, and discourse features
Cognitive Scores	36	1.4	16 (44.4%)	SmrtFactor.computed used for participant grouping and Z-score calculation
Genetic/APOE	10	0.4	4 (40.0%)	Genetic markers; not used in current analysis
Demographics	8	0.3	1 (12.5%)	Age and education used as covariates in Z-score models
Questionnaire/Subjective	5	0.2	0 (0.0%)	Self-report measures; not used in current analysis

Feature categories in the complete BHR-CLASP dataset (N=2,540 features). Categories determined by variable naming conventions and content analysis. Percentages calculated from total feature count. Missingness calculated as proportion of participants with missing data; 2,057 features (81.0%) have >10% missing data. Current analysis focuses on Linguistic/Speech and Acoustic/Waveform features after quality filtering and outlier removal.

Step 2
Missingness Filter

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How Outliers Distort Regression Mathematically

- In simple linear regression:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \qquad \beta_1 = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}$$

- That denominator, the variance of x is where the trouble starts
- Outliers in x create huge deviations from the mean
- These inflate the denominator
- That, in turn, can shrink the slope estimate (β_1), flattening your regression line

Step 3
Outlier Detection

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Example 1/3 Normal Data (No Outlier)

data	
x	y
1	2
2	4
3	6
3.1	6

1 → mean

$$\bar{x} = 2, \quad \bar{y} = 4$$

2 → Compute numerator

$$\sum(x_i - \bar{x})(y_i - \bar{y}) = (1-2)(2-4) + (2-2)(4-4) + (3-2)(6-4)$$
$$= (-1)(-2) + (0)(0) + (1)(2) = 2 + 0 + 2 = 4$$

3 → Compute denominator

$$\sum(x_i - \bar{x})^2 = (1-2)^2 + (2-2)^2 + (3-2)^2 = 1 + 0 + 1 = 2$$

4 → Compute slope

$$\hat{\beta}_1 = \frac{4}{2} = 2$$

Perfect, the slope is 2, every 1 unit increase in x increases y by 2.

regression slope formula :

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

Step 3
Outlier Detection

UCSF

Example 2/3 Normal Data (Add One Outlier)

data	
x	y
1	2
2	4
3	6
3.5	6

1 → mean

$\bar{x} = 2.375 \quad \bar{y} = 4.5$

2 → Compute numerator

Numerator $\sum(x_i - \bar{x})(y_i - \bar{y})$:

$$(-1.375)(-2.5) + (-0.375)(-0.5) + (0.625)(1.5) + (1.125)(1.5)$$
$$= 3.4375 + 0.1875 + 0.9375 = 6.25$$

3 → Compute denominator

Denominator $\sum(x_i - \bar{x})^2$:

$$(-1.375)^2 + (-0.375)^2 + (0.625)^2 + (1.125)^2$$
$$= 1.890625 + 0.140625 + 0.390625 + 1.265625 = 3.6875$$

4 → Compute slope

$$\hat{\beta}_1 = \frac{6.25}{3.6875} \approx 1.69$$

1-IQR-high point shrinks the slope from 2 down to about 1.69
The numerator grew from 4 to 6.25
But the denominator grew from 2 to 3.69
The denominator grew proportionally more, and the slope got smaller.

regression slope formula :

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

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Example 3/3 Normal Data (Add Extreme Outlier)

x	y
1	2
2	4
3	6
5.5	6

1 → mean: $\bar{x} = 2.875 \quad \bar{y} = 4.5$

2 → **Compute numerator**

Numerator $\sum(x_i - \bar{x})(y_i - \bar{y})$:

$$(-1.875)(-2.5) + (-0.875)(-0.5) + (0.125)(1.5) + (2.625)(1.5)$$
$$= 4.6875 + 0.4375 + 0.1875 = 9.25$$

3 → **Compute denominator**

Denominator $\sum(x_i - \bar{x})^2$:

$$(-1.875)^2 + (-0.875)^2 + (0.125)^2 + (2.625)^2$$
$$3.515625 + 0.765625 + 0.015625 + 6.890625 = 11.1875$$

4 → **Compute slope**

$$\hat{\beta}_1 = \frac{9.25}{11.1875} \approx 0.83$$

regression slope formula :

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

as the outlier moves from 1 IQR to 3 IQR out, the denominator (variance of x) explodes, the numerator grows more slowly, and the slope collapses toward zero

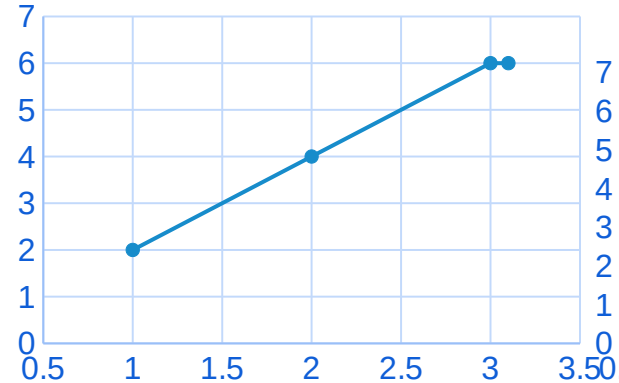
Outlier Distortion Visualized

$$\hat{\beta}_1 = \frac{4}{2} = 2$$

$$\hat{\beta}_1 = \frac{6.25}{3.6875} \approx 1.69$$

$$\hat{\beta}_1 = \frac{9.25}{11.1875} \approx 0.83$$

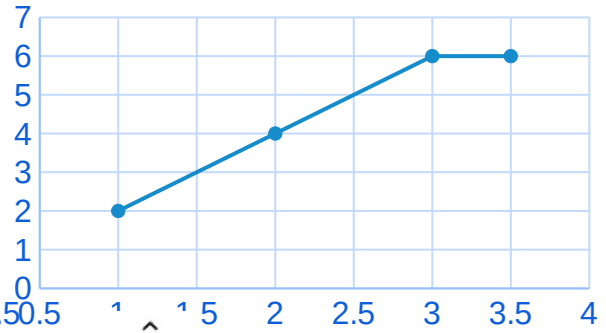
No Outliers



$\hat{\beta}_1 = 2$

x = 3.1 y = 6

1 Mild Outlier (1 IQR)

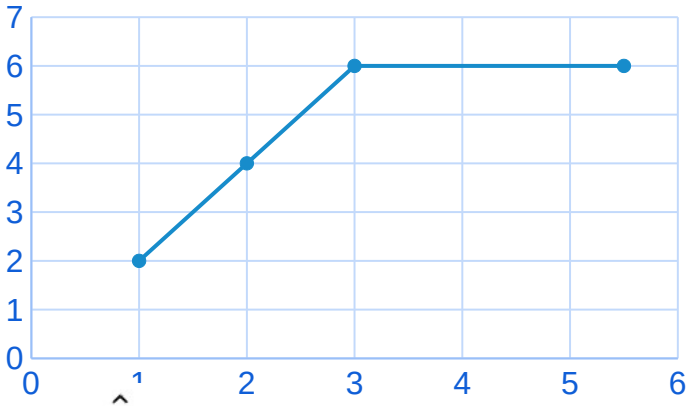


$\hat{\beta}_1 = 1.69$ **Δ** = -.31

x = 3.5
Δ = .4

y = 6

1 Extreme Outlier (3 IQR)



$\hat{\beta}_1 = 0.83$ **Δ** = 1.17

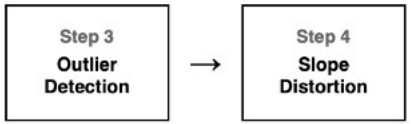
x = 5.5
Δ = 2.4

y = 6

closed-form expression mapping proportion of outliers and distance to slope shrinkage in a simple OLS geometric framework.

$$\hat{\beta}_1(p, c, x_o) = \frac{\beta(1 + p\,c\,x_o^2)}{1 + p(x_o^2 - 1)}$$

Symbol	Meaning
β	The true underlying slope (the population coefficient)
p	Proportion of outliers in the sample
c	Relative bias or directional distortion contributed by each outlier (how much their slope contribution differs from the true slope)
x_o	Scaled leverage or distance of the outliers in predictor-space from the mean (a measure of leverage)



Step 3: Outlier Quantification

IQR-based outlier detection (1.5×IQR and 3×IQR thresholds)

Calculated: Outlier percentages for all variables
- Regular outliers: values beyond Q1 - 1.5×IQR or Q3 + 1.5×IQR
- Extreme outliers: values beyond Q1 - 3×IQR or Q3 + 3×IQR

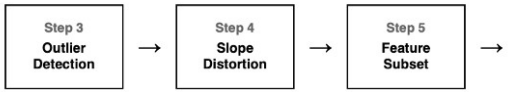
Step 4: Closed-Form Slope Distortion Analysis

Variables with >20% slope distortion removed (Tier 1)

Excluded: 173 variables (mean distortion 47.76%)
Rationale: Outliers inflate variance in denominator of OLS slope estimate, producing >20% bias in regression coefficients

Retained Variables

n = 325 variables (62.6% retention)
Quality distribution: 170 Excellent, 109 Very Good, 46 Other



Imagine you're running a regression to predict exam score based on two variables:

Student	Study Hours (X_1)	Hours Slept (X_2)	Exam Score (Y)
A	2	8	65
B	4	8	75
C	6	8	85
D	8	8	95

- **X_1 = Study Hours** has **high variance**, it changes across students (2, 4, 6, 8)
- **X_2 = Hours Slept** has **zero variance**, everyone wore blue

Basic OLS refresher

- In simple regression, the slope:

$$\beta_1 = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}$$

OLS is literally “how much Y moves when X moves.”

Case 1: variable with variance (Study Hours)

Student	X (Hours Studied)	Y (Score)
A	2	65
B	4	75
C	6	85
D	8	95

Then:

$$\text{Var}(X) = \text{average of } (X_i - \bar{X})^2 = \text{not zero.}$$

Where X_i = data point and $\bar{X}$ = sample mean

And Because X and Y move together

- Then:

$$\text{Cov}(X, Y) > 0 \qquad \beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

- So:

$$\beta_1 = \frac{\text{something}}{\text{something}} = \text{finite number} \rightarrow \text{works!}$$

Case 2: variable with zero variance (Hours Slept)

Student	X (Hours Slept)	Y (Score)
A	8	65
B	8	75
C	8	85
D	8	95

Every $X_i = \bar{X}$ $\beta_1 = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$ $\mathrm{Var}(X) = \frac{1}{n}\sum(X_i - \bar{X})^2 = 0$

That means in the formula:

$$\beta_1 = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} = \frac{0}{0}$$

Regression consequence

In Ordinary Least Squares (OLS), the model looks for relationships between changes in X and changes in Y.

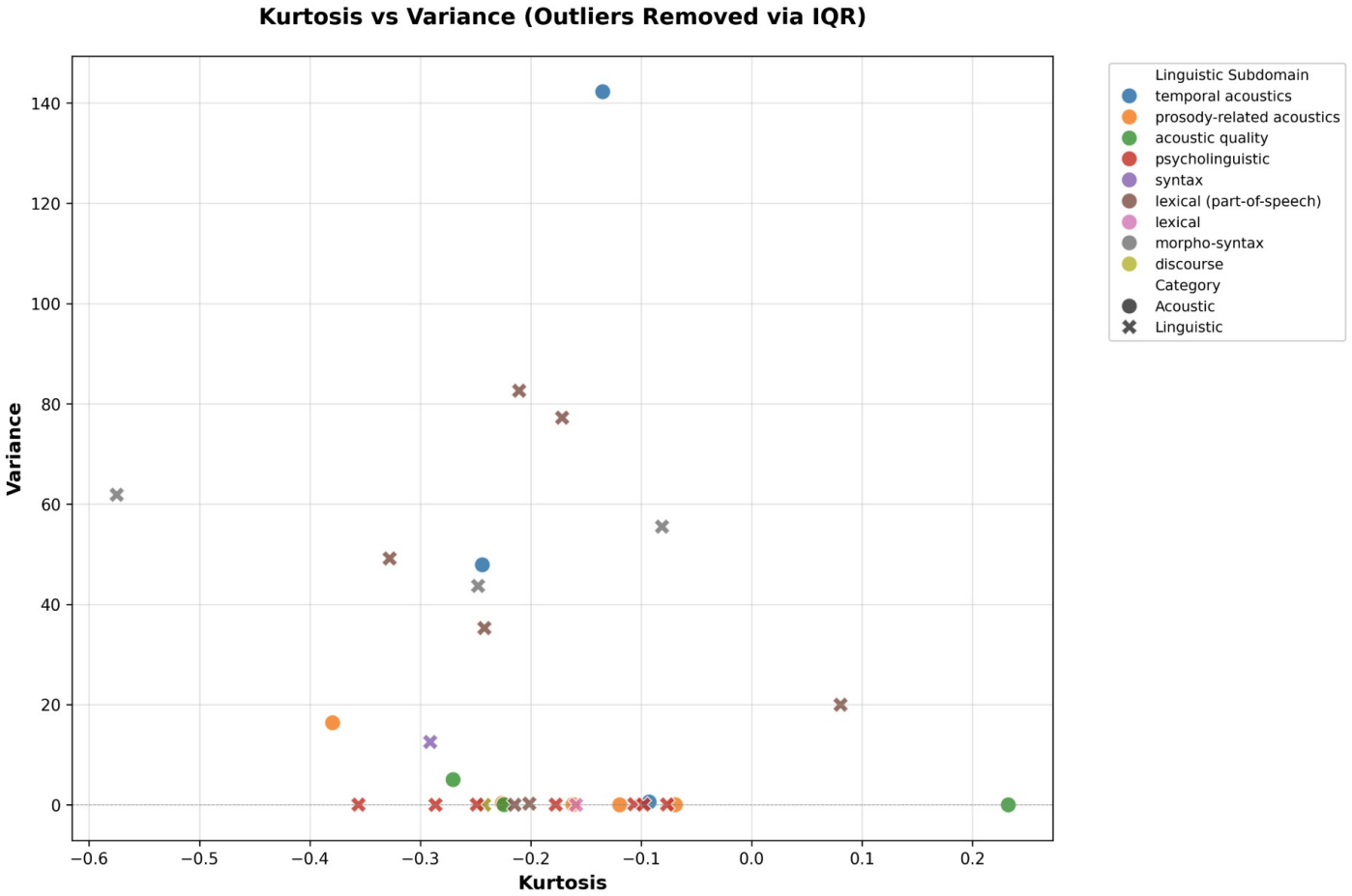
- **Study Hours:** as X_1 increases, Y increases → OLS can estimate a slope (e.g., +5 points per hour).
- **Hours Slept:** since X_2 never changes, OLS can't estimate its slope at all it's mathematically impossible because its variance is **0**.
- If you tried to include hours slept, the model matrix would collapse it would literally divide by zero variance when trying to estimate a coefficient.

Variance is the first gate: no variance, no signal

- If a variable has zero variance, it contributes zero information to the model, it’s a constant.
- Every algorithm, no matter how complex (OLS, random forest, gradient boosting, deep net), internally depends on the existence of variance to create discrimination between cases.

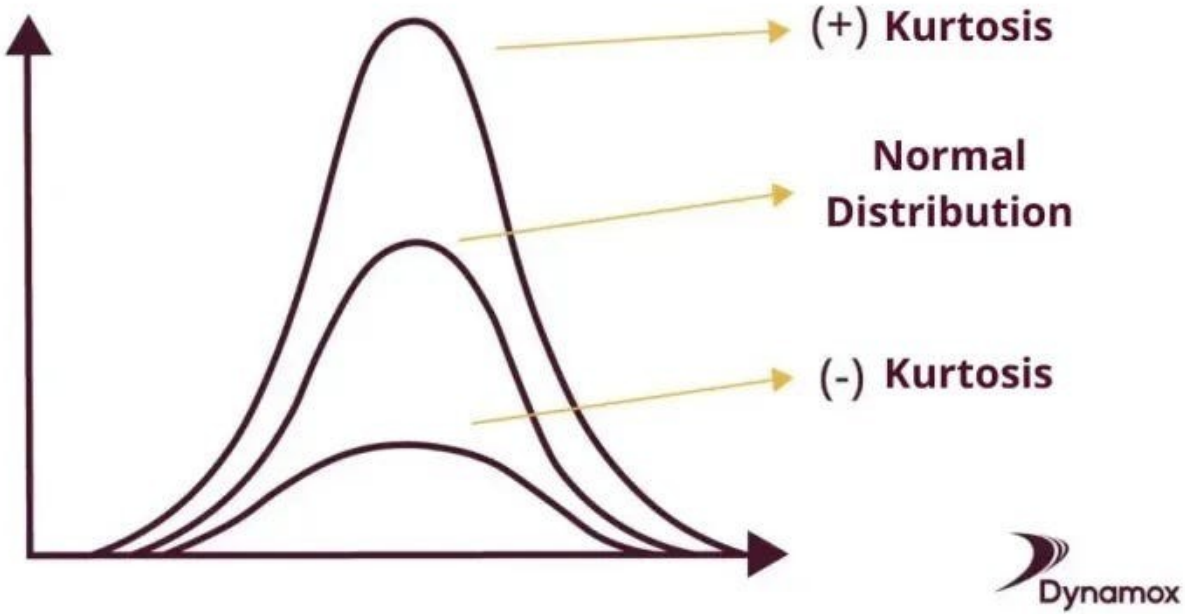
Step 6

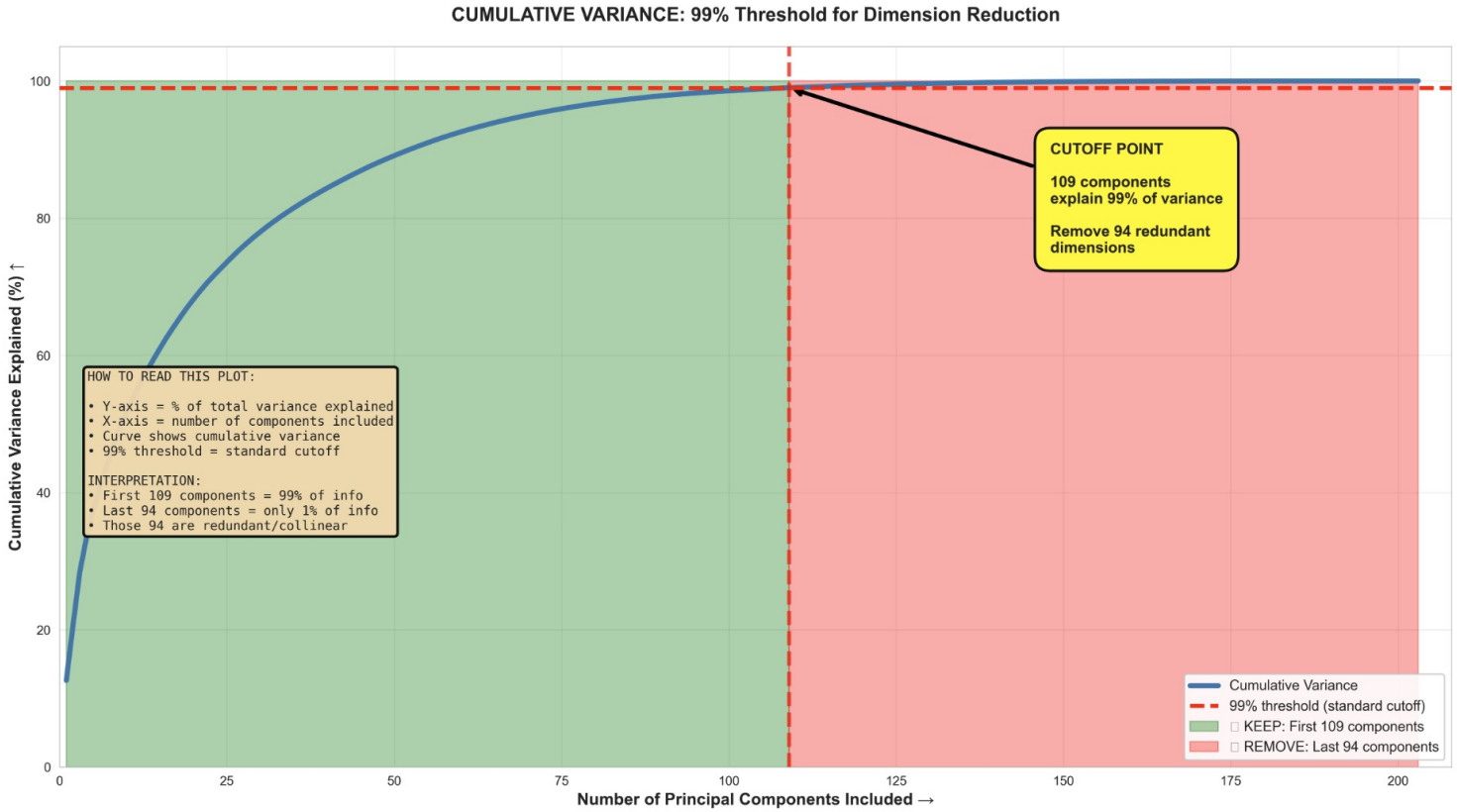
Variance Filter



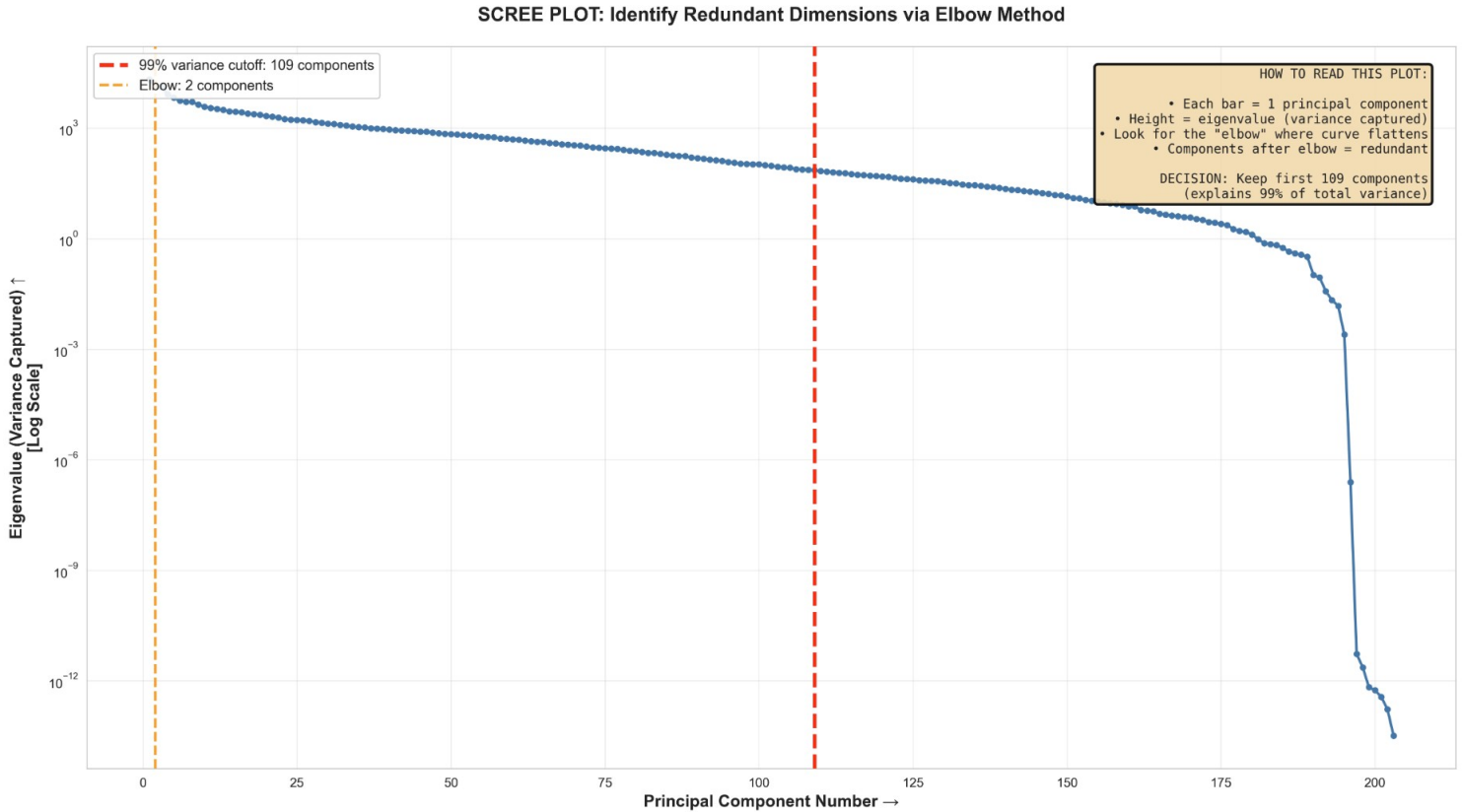
Step 6
Variance Filter

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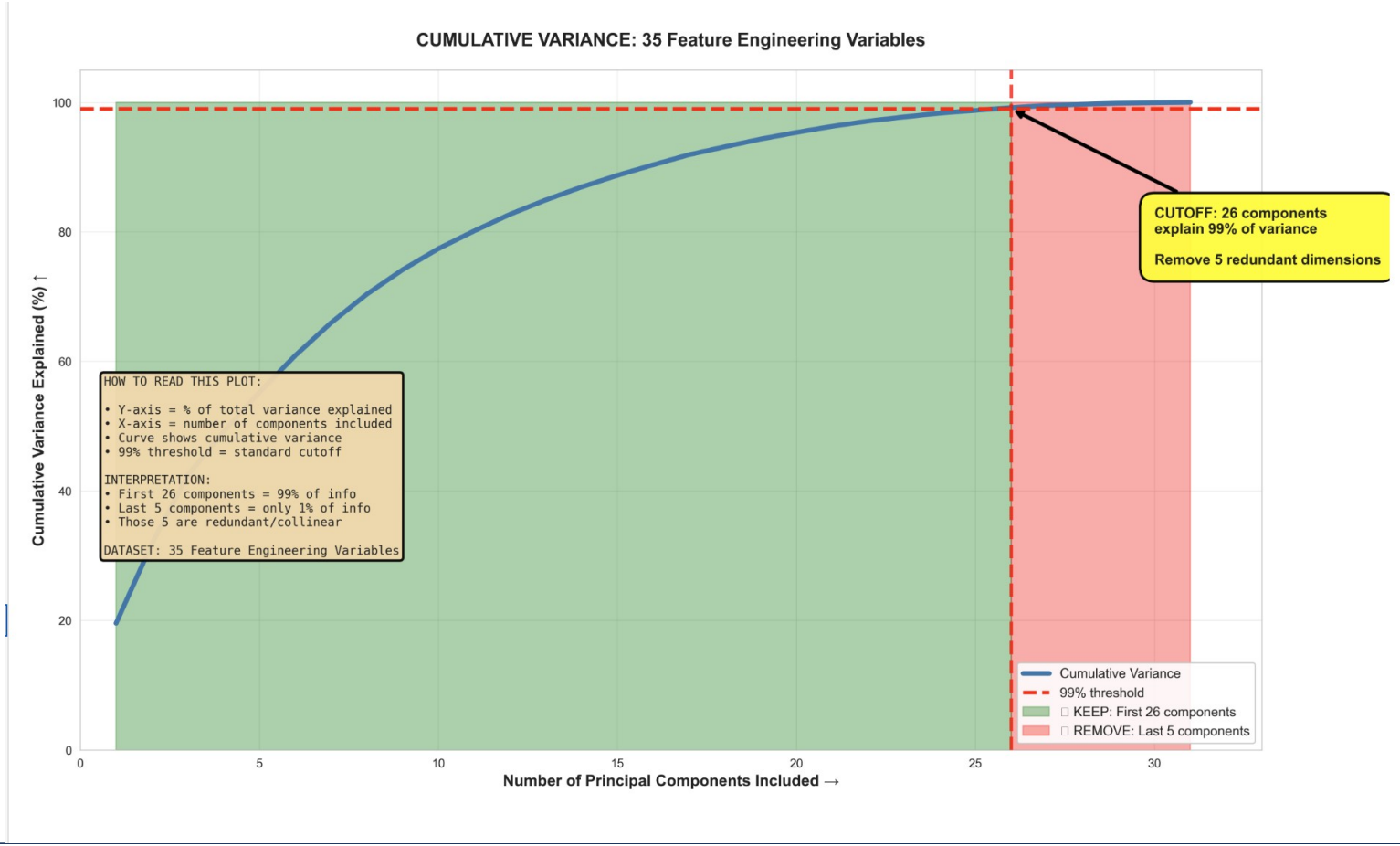

(+) Kurtosis
Normal Distribution
(-) Kurtosis
Dynamox



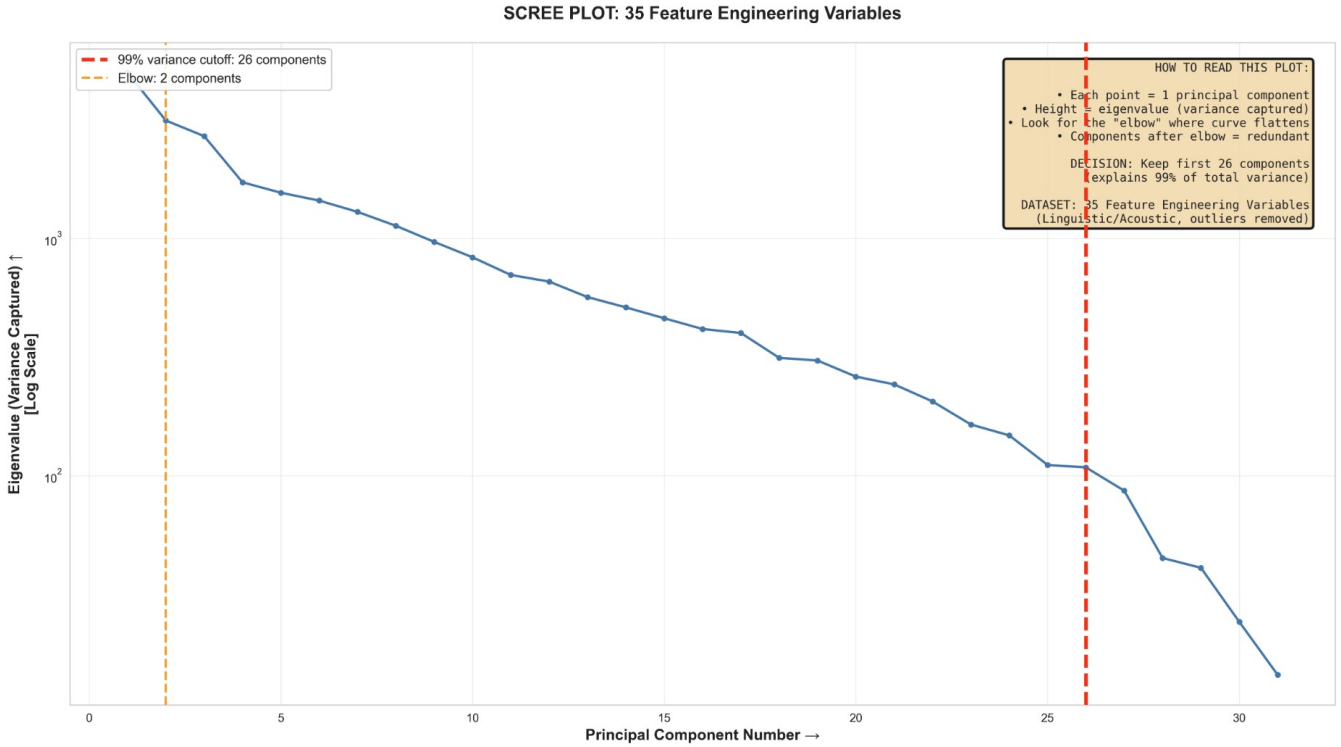
Step 6
Variance Filter



Step 6
Variance Filter



Step 6
Variance Filter



Step 6
Variance Filter

Table: Variance Inflation Factor (VIF) Dimension Reduction via PCA

Step	Method	Variables In	Variables Out	Condition Number (κ)	Rationale
1. Initial	After outlier/missingness removal	325	—	6.54×10^{17}	Catastrophic ill-conditioning; severe multicollinearity
2. PCA Analysis	Eigenvalue decomposition of X^TX	325	—	—	Identified 109 components explaining 99% of variance
3. Dimension Reduction	**Keep top 109 variables by PCA loading**	**109**	**216**	**2.74×10^{16}**	**23.87× improvement in condition number**
4. Feature Engineering Subset	Linguistic/Acoustic only (outliers removed)	32	—	345.94	Final modeling set; well-conditioned ($\kappa < 1000$)

Quasi-Closed-Form Cutoff Criterion: *PCA variance spectrum analysis with 99% cumulative variance threshold. Condition number $\kappa = \lambda_{max}/\lambda_{min}$ quantifies ill-conditioning; high κ indicates near-collinearity. Variables selected based on squared loadings on first 109 principal components. Feature engineering subset (32 variables) represents final analysis-ready data after all quality filters.*

Step 6
Variance Filter

Step 6: Variance Inflation Factor (VIF) Analysis

PCA-based dimension reduction via 99% cumulative variance threshold

Excluded: 94 redundant variables (216 total removed from 325)
Method: Eigenvalue decomposition of X^TX; kept top 109 variables by PCA loading
Result: Condition number κ improved 23.87× ($6.54\times10^{17} \rightarrow 2.74\times10^{16}$)

↓

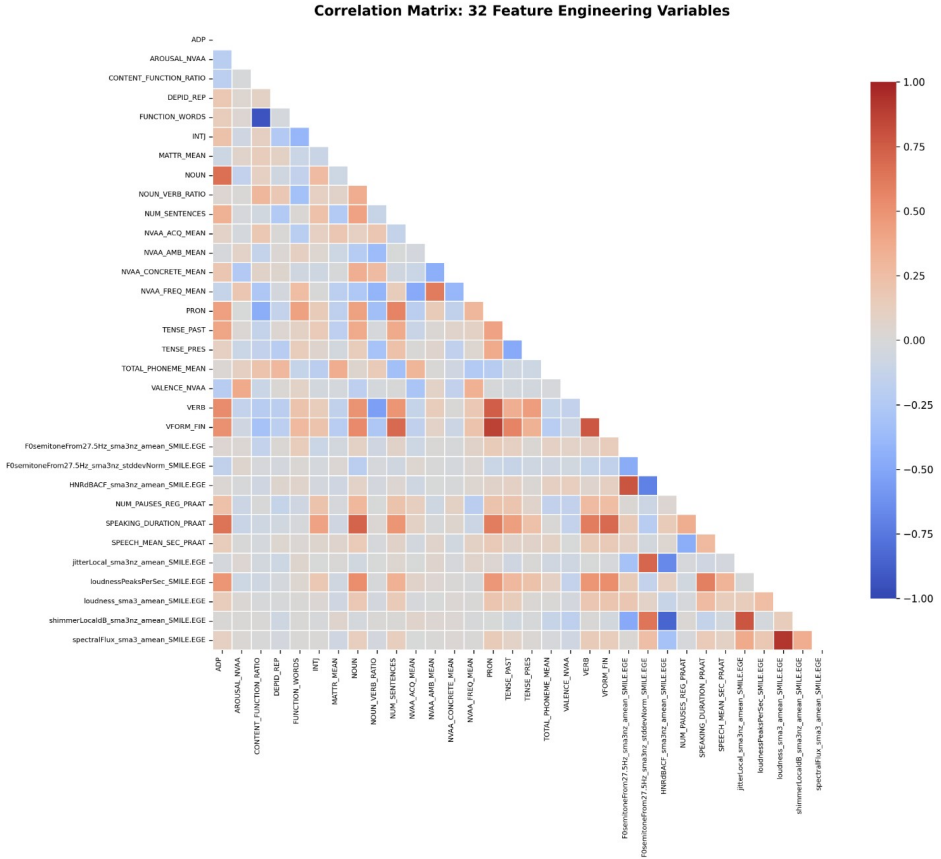
VIF-Reduced Variable Set

n = 109 variables

Quasi-closed-form cutoff: 109 components explain 99% of variance

↓

Step 6
Variance Filter



Step 7
Covariance Reduction

→

Final
Optimized Features

Table High Collinearity Pairs (|r| > 0.7) - 32 Feature Engineering Variables

Variable 1	Variable 2	r	|r|
F1bandwidth_sma3nz_amean_SMILE.EGE	F1bandwidth_sma3nz_stddevNorm_SMILE.EGE	0.988	0.988
F1frequency_sma3nz_stddevNorm_SMILE.EGE	F1frequency_sma3nz_amean_SMILE.EGE	0.984	0.984
F2bandwidth_sma3nz_amean_SMILE.EGE	F2bandwidth_sma3nz_stddevNorm_SMILE.EGE	0.979	0.979
F2frequency_sma3nz_stddevNorm_SMILE.EGE	F2frequency_sma3nz_amean_SMILE.EGE	0.972	0.972
F3bandwidth_sma3nz_amean_SMILE.EGE	F3bandwidth_sma3nz_stddevNorm_SMILE.EGE	0.970	0.970
F3frequency_sma3nz_stddevNorm_SMILE.EGE	F3frequency_sma3nz_amean_SMILE.EGE	0.963	0.963
SPEAKING_DURATION_PRAAT	ORIGIL_DURATION_PRAAT	0.929	0.929
SPEECH_MEAN_SEC_PRAAT	SPEECH_MEDIAN_SEC_PRAAT	0.888	0.888
PAUSE_MEAN_SEC_PRAAT	PAUSE_MEDIAN_SEC_PRAAT	0.867	0.867
ARTICULATION_RATE_PRAAT	SPEAKING_RATE_PRAAT	0.839	0.839
PAUSE_MEAN_SEC_PRAAT	SPEECH_MEAN_SEC_PRAAT	-0.713	0.713

Collinearity analysis on 32 feature engineering variables (Linguistic/Acoustic subset, outliers removed). Pearson correlation coefficient (r) calculated on standardized data. Cutoff threshold: |r| > 0.7 indicates high collinearity. Total pairs analyzed: 496. High collinearity pairs detected: 11. These pairs may require variable selection to reduce multicollinearity in final models.

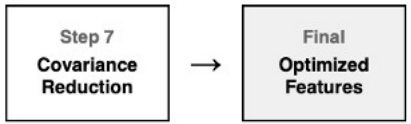


Table High Collinearity Pairs (|r| > 0.7) - 109 VIF-Reduced Variables

Variable 1	Variable 2	r	|r|
CD	NUM	1.000	1.000
pcm_intensity_sma_range_SMILE.EMO	pcm_intensity_sma_max_SMILE.EMO	1.000	1.000
pcm_zcr_sma_max_SMILE.EMO	pcm_zcr_sma_range_SMILE.EMO	0.999	0.999
mfcc4_sma3_amean_SMILE.EGE	mfcc4V_sma3nz_amean_SMILE.EGE	0.952	0.952
F0_INDEX_MYPROS	F0_QUANTILE_75_INDEX_MYPROS	0.947	0.947
ASPECT_PERF	VBN	0.926	0.926
TO	PART	0.882	0.882
F0semitoneFrom27.5Hz_sma3nz_stddevRisingSlope_SMILE.EGE	F0semitoneFrom27.5Hz_sma3nz_meanRisingSlope_SMILE.EGE	0.881	0.881
TOTAL_FAM_PROP_MEAN	TOTAL_FAM_PREV_MEAN	0.879	0.879
pcm_zcr_sma_stddev_SMILE.EMO	pcm_zcr_sma_amean_SMILE.EMO	0.879	0.879
voiceProb_sma_min_SMILE.EMO	voiceProb_sma_range_SMILE.EMO	-0.857	0.857
NVAA_ACQ_MEAN	TOTAL_ACQ_MEAN	0.851	0.851
F2bandwidth_sma3nz_amean_SMILE.EGE	F1bandwidth_sma3nz_amean_SMILE.EGE	0.829	0.829
PERC_CORRECT_VOWEL_MYPROS	FORMANTS_INDEX_MYPROS	-0.828	0.828
INFORM_RATIO	NUM_CONTENT_UNITS	0.826	0.826
DEPID_REP	DEPID	0.824	0.824
AROUSAL_NOUNS	AROUSAL_NVAA	0.806	0.806
NOUNS_ACQ_MEAN	NVAA_ACQ_MEAN	0.803	0.803
voiceProb_sma_stddev_SMILE.EMO	F0semitoneFrom27.5Hz_sma3nz_percentile50.0_SMILE.EGE	0.790	0.790
MOOD_IND	AUX	0.789	0.789
VALENCE_NOUNS	VALENCE_NVAA	0.783	0.783
shimmerLocaldB_sma3nz_amean_SMILE.EGE	jitterLocal_sma3nz_amean_SMILE.EGE	0.774	0.774
F0semitoneFrom27.5Hz_sma3nz_percentile20.0_SMILE.EGE	F0semitoneFrom27.5Hz_sma3nz_percentile50.0_SMILE.EGE	0.762	0.762
F2bandwidth_sma3nz_stddevNorm_SMILE.EGE	F2bandwidth_sma3nz_amean_SMILE.EGE	-0.745	0.745
NOUNS_ACQ_MEAN	TOTAL_ACQ_MEAN	0.744	0.744
equivalentSoundLevel_dBp_SMILE.EGE	pcm_intensity_sma_max_SMILE.EMO	0.732	0.732
equivalentSoundLevel_dBp_SMILE.EGE	pcm_intensity_sma_range_SMILE.EMO	0.732	0.732
slopeV500.1500_sma3nz_amean_SMILE.EGE	slopeUV500.1500_sma3nz_amean_SMILE.EGE	0.715	0.715
jitterLocal_sma3nz_amean_SMILE.EGE	F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE	0.707	0.707
DOMINCE_NOUNS	DOMINCE_NVAA	0.704	0.704

Collinearity analysis on 109 VIF-reduced variables. Pearson correlation coefficient (r) calculated on standardized data. Cutoff threshold: |r| > 0.7 indicates high collinearity. Total pairs analyzed: 5,886. High collinearity pairs detected: 30. These pairs may require further variable selection to reduce multicollinearity in final models.

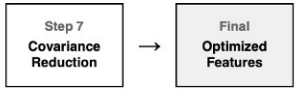


Table Top 20 Regression Models by R^2 (1+N Pairwise Analysis)

Rank	Model	R^2	Adj R^2	β_1	p_1	β_2	p_2	95% CI_2
1	F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + pcm_intensity_sma_range_SMILE.EMO	0.0413	0.0389	-0.095	0.0082	-0.209	<.0001	[-0.28, -0.14]
2	pcm_intensity_sma_range_SMILE.EMO + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE	0.0413	0.0389	-0.209	<.0001	-0.095	0.0082	[-0.17, -0.02]
3	pcm_intensity_sma_range_SMILE.EMO + NVAA_ACQ_MEAN	0.0387	0.0363	-0.183	<.0001	0.076	0.0280	[0.01, 0.14]
4	NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO	0.0387	0.0363	0.076	0.0280	-0.183	<.0001	[-0.25, -0.12]
5	DOMINCE_NVAA + pcm_intensity_sma_range_SMILE.EMO	0.0347	0.0323	0.041	0.2315	-0.182	<.0001	[-0.25, -0.11]
6	pcm_intensity_sma_range_SMILE.EMO + DOMINCE_NVAA	0.0347	0.0323	-0.182	<.0001	0.041	0.2315	[-0.03, 0.11]
7	mfcc4_sma3_amean_SMILE.EGE + pcm_intensity_sma_range_SMILE.EMO	0.0342	0.0319	-0.037	0.2957	-0.188	<.0001	[-0.26, -0.12]
8	pcm_intensity_sma_range_SMILE.EMO + mfcc4_sma3_amean_SMILE.EGE	0.0342	0.0319	-0.188	<.0001	-0.037	0.2957	[-0.11, 0.03]
9	pcm_intensity_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0342	0.0318	-0.180	<.0001	0.036	0.3045	[-0.03, 0.10]
10	voiceProb_sma_stddev_SMILE.EMO + pcm_intensity_sma_range_SMILE.EMO	0.0342	0.0318	0.036	0.3045	-0.180	<.0001	[-0.25, -0.11]
11	DEPID_REP + pcm_intensity_sma_range_SMILE.EMO	0.0338	0.0314	-0.029	0.4004	-0.181	<.0001	[-0.25, -0.11]
12	pcm_intensity_sma_range_SMILE.EMO + DEPID_REP	0.0338	0.0314	-0.181	<.0001	-0.029	0.4004	[-0.10, 0.04]
13	jitterLocal_sma3nz_amean_SMILE.EGE + pcm_intensity_sma_range_SMILE.EMO	0.0335	0.0311	-0.025	0.5039	-0.191	<.0001	[-0.26, -0.12]
14	pcm_intensity_sma_range_SMILE.EMO + jitterLocal_sma3nz_amean_SMILE.EGE	0.0335	0.0311	-0.191	<.0001	-0.025	0.5039	[-0.10, 0.05]
15	pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO	0.0335	0.0311	-0.186	<.0001	0.023	0.5115	[-0.05, 0.09]
16	pcm_zcr_sma_amean_SMILE.EMO + pcm_intensity_sma_range_SMILE.EMO	0.0335	0.0311	0.023	0.5115	-0.186	<.0001	[-0.26, -0.12]
17	shimmerLocaldB_sma3nz_amean_SMILE.EGE + pcm_intensity_sma_range_SMILE.EMO	0.0334	0.0310	-0.022	0.5331	-0.187	<.0001	[-0.26, -0.12]
18	pcm_intensity_sma_range_SMILE.EMO + shimmerLocaldB_sma3nz_amean_SMILE.EGE	0.0334	0.0310	-0.187	<.0001	-0.022	0.5331	[-0.09, 0.05]
19	AROUSAL_NVAA + pcm_intensity_sma_range_SMILE.EMO	0.0333	0.0309	0.020	0.5622	-0.182	<.0001	[-0.25, -0.11]
20	pcm_intensity_sma_range_SMILE.EMO + AROUSAL_NVAA	0.0333	0.0309	-0.182	<.0001	0.020	0.5622	[-0.05, 0.09]

Top 20 regression models ranked by R^2. N = 809. Models include single variables and pairwise combinations (1+N). β_1 = coefficient for base variable, β_2 = coefficient for added variable (if applicable), p = p-value, CI = 95% confidence interval. All variables standardized.

Table Top 20 Models by R² (All Possible Combinations)

Rank	N	Variables	R²	Adj R²
1	13	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0563	0.0409
2	12	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0563	0.0421
3	12	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0563	0.0420
4	11	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_stddev_SMILE.EMO	0.0563	0.0432
5	12	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO	0.0562	0.0420
6	11	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO	0.0562	0.0432
7	11	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO	0.0562	0.0431
8	10	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO	0.0562	0.0443
9	12	AROUSAL_NVAA + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0558	0.0416
10	11	AROUSAL_NVAA + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0558	0.0428
11	11	AROUSAL_NVAA + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0558	0.0428
12	10	AROUSAL_NVAA + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0558	0.0440
13	12	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0558	0.0416
14	11	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0558	0.0428
15	11	AROUSAL_NVAA + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO	0.0557	0.0427
16	10	AROUSAL_NVAA + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO	0.0557	0.0439
17	11	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + pcm_zcr_sma_amean_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0557	0.0427
18	10	AROUSAL_NVAA + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + shimmerLocaldB_sma3nz_amean_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO	0.0557	0.0439
19	9	AROUSAL_NVAA + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + mfcc4_sma3_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO	0.0557	0.0451
20	10	AROUSAL_NVAA + DEPID_REP + DOMINCE_NVAA + F0semitoneFrom27.5Hz_sma3nz_stddevNorm_SMILE.EGE + jitterLocal_sma3nz_amean_SMILE.EGE + NVAA_ACQ_MEAN + pcm_intensity_sma_range_SMILE.EMO + VALENCE_NVAA + voiceProb_sma_range_SMILE.EMO + voiceProb_sma_stddev_SMILE.EMO	0.0557	0.0439

Top 20 models from all possible combinations of 13 variables. N = 809. Total combinations tested: 8,191. All variables standardized.

Table Age-Stratified Regression Results (All 13 Variables)

Age Group	N	R^2	Adj R^2
<65	428	0.0415	0.0114
65-74	283	0.1476	0.1064
75-84	89	0.1357	-0.0141

Table Enhanced Regression Models

Model	R^2
Full enhanced model	0.4028
13 vars + age	0.3855
13 vars + shimmerLocaldB_sma3nz_amean_SMILE.EGE × pcm_zcr_sma_amean_SMILE.EMO	0.0767
13 vars + jitterLocal_sma3nz_amean_SMILE.EGE × pcm_zcr_sma_amean_SMILE.EMO	0.0682
13 vars + shimmerLocaldB_sma3nz_amean_SMILE.EGE × mfcc4_sma3_amean_SMILE.EGE	0.0674
13 vars + AROUSAL_NVAA × voiceProb_sma_range_SMILE.EMO	0.0655
13 vars + jitterLocal_sma3nz_amean_SMILE.EGE × mfcc4_sma3_amean_SMILE.EGE	0.0651
13 vars + 3 squared terms	0.0622
Baseline (13 vars)	0.0563

N = 809. Enhanced models include age, squared terms, and interaction terms.

Table 1. Acoustic Feature Comparison Between 16 kHz and 44.1 kHz Recording Groups

Forensic analysis of waveform characteristics reveals systematic differences between recordings sampled at 16 kHz (n=544) and 44.1 kHz (n=251). Features were extracted using frame-based analysis (25 ms frames, 10 ms hop) from raw audio waveforms. Values represent mean ± standard deviation across all recordings in each group. RMS = root-mean-square; SD = standard deviation. Spectral centroid indicates the "center of mass" of the spectrum (brightness); spectral flatness measures how noise-like (vs. tone-like) the signal is; zero-crossing rate indicates signal noisiness; spectral rolloff indicates the frequency below which 85% of spectral energy is contained. Large differences in spectral centroid (1527 Hz vs 3432 Hz) and zero-crossing rate (0.125 vs 0.084) indicate fundamentally different recording pipelines, not merely different sampling rates. This represents a major batch effect that may confound downstream linguistic and acoustic analyses.

Acoustic Feature	Description	16 kHz Group (n=544)	44.1 kHz Group (n=251)
RMS Energy (Mean)	*Mean root-mean-square energy across frames*	0.0483 ± 0.0250	0.0382 ± 0.0199
RMS Energy (SD)	*Standard deviation of RMS energy*	0.0476 ± 0.0241	0.0377 ± 0.0188
Zero-Crossing Rate (Mean)	*Mean zero-crossing rate (noisiness indicator)*	0.1245 ± 0.0293	0.0845 ± 0.0451
Spectral Centroid (Mean, Hz)	*Mean spectral centroid (brightness)*	1526.6502 ± 257.8215	3432.1395 ± 1349.2960
Spectral Flatness (Mean)	*Mean spectral flatness (noise-like vs tone-like)*	0.2812 ± 0.0555	0.2191 ± 0.0881
Spectral Rolloff (Mean, Hz)	*Mean spectral rolloff (frequency content)*	2861.1438 ± 485.3471	6642.5396 ± 2764.0339
Amplitude Entropy	*Shannon entropy of amplitude distribution*	2.2014 ± 0.4493	2.0131 ± 0.4538
Silence Fraction	*Proportion of frames with RMS < 0.01*	0.3381 ± 0.1363	0.3844 ± 0.1461
Clipping Fraction	*Proportion of samples near full scale*	0.0000 ± 0.0000	0.0000 ± 0.0000